

# The Empirical Case for Gaussian Renormalization as Default

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## Abstract

A population’s full-menu choice shares do not determine its choices from smaller menus. Luce’s axiom supplies one rule connecting them, renormalization, which leaves the odds between survivors untouched; an independent Gaussian race supplies another, which makes the survivors less lopsided. We observe that the two are calibrated from exactly the same shares, so they can be compared as *restriction maps* with nothing fitted to the smaller menus, and we score them out of sample. The Gaussian map is better on an experiment in which subjects chose separately from all thirty-one subsets of five goods, and on twelve of twelve archival ranking collections. Both results survive a null in which the axiom holds by construction, which is necessary because contracting a noisily estimated share vector is also shrinkage. The recommendation is narrow: when mapping aggregate full-menu shares to restricted-menu shares, make the race the default.

## 1 Two restriction maps

For a population and a menu  $S$ , let  $p_i$  be the share choosing or first-ranking item  $i$ . Luce’s axiom [5] assigns a worth  $u(i)$  with

$$p_i = \frac{u(i)}{\sum_{j \in S} u(j)}, \quad (1)$$

so the shares determine the worths up to scale, and the prediction for any subset  $T \subseteq S$  follows by changing the denominator alone. Odds between survivors are untouched: for  $i, j \in T$  the share preferring  $i$  is  $p_i/(p_i + p_j)$  when  $T$  is the pair. This is Independence of Irrelevant Alternatives.

Thurstone [8] connects the same two quantities through a contest. Item  $i$  carries a location  $a_i$ , draws a performance  $X_i \sim N(a_i, 1)$  on each occasion, and the highest draw is chosen, so

$$p_i = \int \varphi(x - a_i) \prod_{k \neq i} \Phi(x - a_k) dx. \quad (2)$$

The integral is one-dimensional for any number of alternatives, since conditioning on  $X_i = x$  makes the other events independent, and it is inverted numerically by a lattice method [2]; locations are identified up to a common translation, fixed here by  $\sum_i a_i = 0$ . Prediction for a subset  $T$  means re-running the contest among  $T$  alone. For a pair this is one normal evaluation,

$$P(X_i > X_j) = \Phi\left(\frac{a_i - a_j}{\sqrt{2}}\right), \quad (3)$$

and it is less lopsided than  $p_i/(p_i + p_j)$ : winning a large field needs an unusually good draw, and a favourite’s edge is partly an edge in producing extremes, so that edge counts for less when one rival remains.

Both rules are calibrated to the same  $|S| - 1$  free shares and neither sees any observation from a smaller menu. The comparison is therefore between two zero-extra-parameter maps, not between two fitted models. That is worth stating plainly, because it also bounds what the comparison can show. A fitted Plackett–Luce likelihood uses the whole ranking and a fitted Bradley–Terry model uses the pairwise outcomes directly; neither is tested here. What is tested is a cross-moment implication of the homogeneous Luce model, namely whether its full-menu shares correctly determine its restricted-menu shares.

The direction of the disagreement is a theorem. With the highest draw winning, let  $r = f/F$  be the reverse hazard of the common noise. For nested menus  $T \subset S$  and  $i, j \in T$  with  $a_i > a_j$ , if  $\log r$  is concave then the odds of  $i$  against  $j$  are at least as large in  $S$  as in  $T$ , strictly so when the concavity is strict and at least one contestant is actually removed. The argument is a covariance inequality between two increasing functions.

Two laws sit at the ends of that condition. The Gaussian satisfies it strictly, since  $\frac{d^2}{dx^2} \log r(x) = -\text{Var}(Z \mid Z < x) < 0$ . The Gumbel has  $r(x) = s^{-1}e^{-x/s}$ , so  $\log r$  is affine, the inequality becomes an equality, and renormalization is recovered exactly. That is the sufficiency half of Yellott’s theorem [9]. The axiom is a boundary of the class, not a rival to it.

We use the proposition for the sign and nothing else. It orders no magnitudes, and contraction is not a property of a noise law: it depends on the share configuration, and laws recalibrated to the same shares reorder between configurations. Heavier tails do not reliably contract less.

## 2 Choices from real menus

Twelve of the thirteen collections used here are complete rankings, from which subset choice is read off rather than observed. One is not, and it comes first.

Costa-Gomes, Cueva, Gerasimou and Tejiscak [1] had each subject choose separately from all thirty-one subsets of five consumer goods, so restriction is observed rather than imputed. The archive names the goods only by two-letter codes and we describe them no more precisely. This is the cleanest available test of the question above, and it is the primary result.

Subjects are split into five folds. On the training subjects, shares come only from the full five-item menu; both rules are calibrated to those shares and nothing else; each then predicts every menu of size two, three and four; and we score the choices actually made by held-out subjects. Intervals resample subjects and repeat the whole procedure, calibration included. Some subjects may defer, so the forced-choice subgroup is reported separately: those subjects cannot defer, and no coding decision about deferral enters that row.

| Subjects           | $n$ | Choices | Renorm. | Race   | Gain                       |
|--------------------|-----|---------|---------|--------|----------------------------|
| All                | 364 | 8,239   | 0.9263  | 0.9164 | +0.0100 [+0.0048, +0.0246] |
| Forced choice only | 137 | 3,425   | 0.9112  | 0.8846 | +0.0265 [+0.0116, +0.0700] |

Table 1: Held-out log loss per choice on menus of size two to four, with both rules calibrated only to training-fold full-menu shares under a common add- $\alpha$  convention,  $\alpha = 1/2$ . Shares rest on 268 full-menu choices for all subjects and 109 for the forced-choice subgroup. Intervals are subject bootstraps that recompute folds, shares, calibration and predictions.

The Gaussian rule is better on both, and better by more where deferral cannot interfere. Section 4 shows the gain is not an artefact of estimating shares from a few hundred observations.

### 3 Choices induced from rankings

Twelve archival collections have human respondents and rankings, or ratings fine enough to yield them. Two rank-order card tasks from the General Social Survey ask what a child should learn and what matters in a job. Three independent samples of Jester users rate the same ten jokes. Sushi rankings come from a Japanese web survey, film rankings are induced from Netflix ratings, and dots and puzzles are perceptual discrimination tasks. The rest are rankings of seven sports by participation preference, ten occupations by perceived prestige, and four political goals from a five-nation survey. The thirteen collections comprise 147,719 responses, of which the menu experiment contributes 373 subjects; the twelve ranking collections contribute 147,346. These are not thirteen independent experiments, since three are Jester files, two are General Social Survey items, and dots and puzzles are companion tasks.

The protocol matches Section 2 with respondents in place of subjects, and the outcome for a subset is the respondent’s highest-ranked member of it.

| Dataset               | $n$   | $K$ | Subsets | Renorm. | Race   | Gain                       |
|-----------------------|-------|-----|---------|---------|--------|----------------------------|
| Occupational prestige | 143   | 10  | 1,013   | 1.0173  | 0.9783 | +0.0390 [+0.0227, +0.0555] |
| Sushi                 | 5,000 | 10  | 1,013   | 1.3724  | 1.3613 | +0.0111 [+0.0094, +0.0128] |
| Political goals       | 2,262 | 4   | 11      | 0.8835  | 0.8766 | +0.0069 [+0.0059, +0.0078] |
| GSS socialization     | 5,000 | 5   | 26      | 0.7984  | 0.7929 | +0.0055 [+0.0041, +0.0070] |
| Jester file 3         | 5,000 | 10  | 1,013   | 1.5189  | 1.5165 | +0.0024 [+0.0015, +0.0032] |
| Jester file 1         | 5,000 | 10  | 1,013   | 1.5287  | 1.5267 | +0.0020 [+0.0012, +0.0027] |
| Jester file 2         | 5,000 | 10  | 1,013   | 1.5247  | 1.5230 | +0.0018 [+0.0010, +0.0025] |
| GSS job values        | 5,000 | 5   | 26      | 0.8593  | 0.8576 | +0.0017 [+0.0005, +0.0030] |
| Dots                  | 3,183 | 4   | 11      | 0.8285  | 0.8270 | +0.0015 [+0.0005, +0.0025] |
| Netflix               | 4,379 | 3   | 4       | 0.7932  | 0.7922 | +0.0009 [+0.0007, +0.0012] |
| Puzzles               | 3,180 | 4   | 11      | 0.8180  | 0.8173 | +0.0007 [-0.0003, +0.0017] |
| Sports participation  | 130   | 7   | 120     | 1.2579  | 1.2528 | +0.0050 [+0.0017, +0.0087] |

Table 2: Held-out log loss per prediction over every subset of size two or more. Datasets are capped at five thousand respondents for runtime. The race is lower in twelve of twelve and the interval excludes zero in eleven. Occupational prestige is scorable here only because the shared smoothing removes a zero training cell. Sports participation sits below the rule: its gain is real but Section 4 shows it carries no evidence about the restriction map. Intervals resample respondent losses with the fitted training models held fixed, so unlike Table 1 they omit calibration uncertainty and are too narrow, most seriously in the two smallest rows.

Averaging over every subset weights cardinalities by how many there are, so intermediate menus dominate: for  $K = 10$  only 45 of 1,013 subsets are pairs, about four per cent of the weight. Those middling menus are where least has been removed and least is at stake. Splitting by menu size shows the pairwise gain exceeding the aggregate in eleven of twelve datasets, by factors from 0.7 to 3.7 with median 1.8: Sushi is +0.0412 at  $|T| = 2$  against +0.0111 aggregate, and GSS socialization +0.0130 against +0.0055. Occupational prestige is the exception, peaking at  $|T| = 4$ . At  $|T| = K$  nothing has been removed, the two rules must agree, and the measured

gain is zero to four decimals in every dataset, which is the check that the pipeline is sound. We keep the all-subsets aggregate as the headline because it fixes the estimand in advance rather than selecting a cardinality after the fact.

## 4 Shrinkage

A positive gain has two possible sources and log loss cannot tell them apart. The race may be closer to how the population chooses. Or the race may be wrong about that and win anyway, because contracting a noisily estimated share vector is shrinkage, and shrinkage lowers log loss when the input is over-confident. The second mechanism needs no departure from the axiom, so properness of the score licenses nothing here: a proper score says the true probability vector is optimal, not that a plug-in estimate of it beats a biased transformation of the same estimate.

The remedy is to build a population where the axiom holds by construction and run the identical procedure on it. Taking the observed full-menu shares as worths, choices or complete rankings are generated from that exact process, and everything is repeated. Rankings are drawn by sorting  $\log u_i + G_i$  for independent standard Gumbel  $G_i$ , which is the Plackett–Luce ranking law; for  $K \geq 4$  this is a particular joint law consistent with those subset marginals rather than the only one, so the null is specifically a Plackett–Luce ranking null.

| Dataset                        | Observed | Null median | Excess  | $p$          |
|--------------------------------|----------|-------------|---------|--------------|
| Menu experiment, forced choice | +0.0265  | −0.0041     | +0.0306 | 0.010        |
| Menu experiment, all subjects  | +0.0100  | −0.0023     | +0.0123 | 0.015        |
| Occupational prestige          | +0.0390  | −0.0227     | +0.0617 | $\leq 0.005$ |
| Sushi                          | +0.0111  | −0.0062     | +0.0174 | $\leq 0.005$ |
| GSS socialization              | +0.0055  | −0.0066     | +0.0121 | $\leq 0.005$ |
| Political goals                | +0.0069  | −0.0012     | +0.0081 | $\leq 0.005$ |
| GSS job values                 | +0.0017  | −0.0047     | +0.0065 | $\leq 0.005$ |
| Jester file 3                  | +0.0024  | −0.0015     | +0.0039 | $\leq 0.005$ |
| Jester file 1                  | +0.0020  | −0.0014     | +0.0034 | $\leq 0.005$ |
| Jester file 2                  | +0.0018  | −0.0014     | +0.0032 | $\leq 0.005$ |
| Dots                           | +0.0015  | −0.0014     | +0.0029 | $\leq 0.005$ |
| Puzzles                        | +0.0007  | −0.0011     | +0.0019 | $\leq 0.005$ |
| Netflix                        | +0.0009  | −0.0001     | +0.0010 | $\leq 0.005$ |
| Sports participation           | +0.0050  | +0.0051     | −0.0000 | 0.51         |

Table 3: The same gain when the axiom holds by construction. Two hundred replicates per row, so  $p$  is floored at  $1/201$ ; the menu experiment uses its own 373 subjects and menus. Wherever shares rest on more than about two thousand observations the null is negative, because contraction is then a bias rather than useful insurance, so comparing an observed gain with zero understates it.

The shrinkage mechanism is real and it accounts for the entire gain in the smallest dataset. Sports participation, with 130 respondents, sits exactly on its null and is excluded from every claim here. Everywhere else the mechanism runs against the race, and the excess over the null exceeds the raw gain in all twelve remaining rows. It also rescues puzzles, whose raw interval in Table 2 covers zero while its excess over the null does not. The menu experiment clears the null on both readings, which is the result the paper rests on, since it is the only one where restriction

is observed.

## 5 The benchmark gap

An absolute gain of a few thousandths of a nat is a valid predictive effect size but a poor guide to how much was on offer, because both rules receive the same shares and the shares carry nearly all the predictive content of a restricted menu. To scale it we also score a smoothed subset-frequency forecast, the training-fold frequency with which each item is top within each subset, which is  $|T| - 1$  free probabilities per subset.

This benchmark is not an oracle ceiling, and treating it as one would overstate the ratio. It is itself estimated, so its expected held-out loss exceeds the oracle conditional entropy by roughly  $(m - 1)/2n$  for an  $m$ -outcome menu and  $n$  training observations, and a regularized or structured forecast can beat it out of sample. Applying that first-order correction, the Gaussian race closes a median 35 per cent of the empirical benchmark gap, from 16 per cent for sports participation to 58 per cent for dots. Uncorrected the median is 39 per cent. Either way the reading is that a rule with nothing fitted to restricted menus recovers a third or more of what a per-subset frequency estimator achieves, while renormalization recovers none of it, being the unique odds-preserving map. Denominators as small as 0.002 make these ratios unstable, and they are offered as context for the absolute gains rather than as a headline.

## 6 Scope

The practical recommendation is narrow. When the task is to infer restricted-menu population shares from full-menu population shares, the Gaussian race should be the default in place of renormalization. It costs one extra one-dimensional integral, needs no additional data, and was better on every collection examined here except the one too small to discriminate. In the single experiment where restriction is actually observed rather than induced from rankings, it is better by +0.0265 nats per choice among subjects who cannot defer.

Ten of the twelve ranking collections carry an assumption the menu experiment does not. A respondent's ranking imposes one ordering on every subset by construction, so no ranking dataset can show a person choosing  $i$  over  $j$  from one menu and  $j$  over  $i$  from another, and none reveals whether that person would choose as their ranking implies when handed two options. Reading restriction off a ranking assumes they would. Section 2 does not.

Every quantity here is a population quantity. A population of people who each choose by Equation (1), with worths differing between people, generally does not satisfy the axiom in aggregate, which is why mixed logit is useful and why its probabilities are integrals over logit probabilities rather than logit probabilities [7]. Aggregate failure of the axiom is therefore not evidence that individuals violate it. Heterogeneous Luce choosers reproduce aggregate contraction. What is rejected is a representative-agent map.

The experiment compares two restriction maps, not two fitted likelihoods. A fitted Plackett–Luce model uses the whole ranking and a fitted Bradley–Terry model uses the pairwise outcomes directly. Neither was estimated here. Recommendations about ranking or paired-comparison likelihoods therefore have no support in these tables.

The score is a sum of marginal scores for top-item-within-subset outcomes. It is proper for those marginals and not for the ranking distribution. With  $K = 4$  the scored marginals carry at

most 17 degrees of freedom against 23 for a distribution over rankings, so distinct ranking laws can agree on everything scored here.

Nesting is not costlessness. The independent-noise class contains the Gumbel point, so permitting the class excludes no Luce population, but Table 3 shows that choosing the Gaussian point is a separate statistical decision which loses when the axiom holds and the shares are well estimated. Neither map dominates the other. Each owns a regime, and the border is drawn by how precisely the full-menu shares are known.

A tempting shortcut fails. Pricing ordered outcomes by removing the winner and re-running looks like the race’s own prediction, and it is not: against the calibrated race’s ordering law it misprices individual pairs and triples by up to a factor of 3.4. We leave to future work the comparison of the exact Gaussian ordering law against Plackett–Luce ranking probabilities on complete-ranking data.

## Data and code

Sushi rankings come from Kamishima [4]; Jester ratings from the Eigentaste project [3]; the General Social Survey cumulative file from NORC; the perceptual, leisure, prestige, film and political-goal rankings via PrefLib [6] and the CRAN packages PLMIX and ConsRank; the menu-choice data from the replication archive of [1]. All are public. Analysis code and every input needed to reproduce the tables are under `research/human` in `github.com/microprediction/winning`, the inputs as column extracts rather than raw archives.

Two storage conventions must be told apart when reproducing this. BayesMallows, ConsRank and PerMallows store ranks, so element  $j$  holds the rank of alternative  $j$ , while PLMIX stores orderings, so the first element names the favourite. Reading one as the other transposes the data silently and raises no error. We established which is which by cross-validation: the political goals data ships in two of these packages and they agree only when each is read under its own convention.

## References

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